

Department of Mathematical Sciences
Faculty of Science
Sa'adu Zungur University (Bauchi State University), Gadau

End of First Semester Examination — 2023/2024 Session
Course Code & Title: MTH230 — Mathematical Methods I
Credit Units: 3 Units Time Allowed: 3 Hours
Instruction: Answer Any Five (5) Questions

SECTION A — Attempt Any Three Questions

Question 1

(a) State Rolle's Theorem.

(b) Verify Rolle's Theorem on:

$$f(x) = \sin x / e^x \text{ on } [0, \pi]$$

(c) Given that $\mathbf{R}(t) = (3 \sin t)\mathbf{i} + (3 \cos t)\mathbf{j} + (4t)\mathbf{k}$, find:

(i) $\mathbf{V} / |\mathbf{V}|$

x	y	z
x	y	z
x	y	z

(ii) $\nabla \times \mathbf{a} / |\nabla \times \mathbf{a}|$

(iii) Radius of curvature = $|\nabla \times \mathbf{a}| / |\nabla \times \mathbf{a}|^3$

Question 2

(a) Construct a Taylor's series expansion of $f(x) = \sin x$, given $x = \pi$.

(b) Give the Taylor's series expansion of $f(x) = x^4 + x - 2$ at $x = 2$.

Question 3

(a) Show that the Maclaurin series:

$$\sqrt{(1+x)/(1-x)} = x + x^3/3 + x^5/5 + x^7/7 + \dots$$

(b) Given that $f(x) = \sqrt{x}$, $a = 4$, determine the error at P_2 at point $[3, 5]$.

SECTION B — Attempt Any Two Questions

Question 4 (Section B — Q5)

(a) If $z = e^{r \cos \theta} \cos(r \sin \theta)$, find:

(i) $\partial z / \partial x$

(ii) $\partial z / \partial \theta$

(b) Given that $f = \sin^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{2})$, show that:

$$b \cdot \partial f / \partial x + a \cdot \partial f / \partial y = 2abf$$

Question 5 (Section B — Q6)

(a) Given that $f = e^{ax + by} \cdot f(ax - by)$, show that:

$$b \cdot \partial f / \partial x + a \cdot \partial f / \partial y = 2abf$$

(b) If $z = (1 - 2xy + y^2)^{-1/2}$, show that:

$$x \cdot \partial z / \partial x - y \cdot \partial z / \partial y = y^2 z^3$$

Question 6 (Section B — Q7)

(a) Obtain the reduction formula of:

$$I = \int_0^{\pi/2} \sin^n x \, dx$$

(b) Hence evaluate:

$$I = \int_0^{\pi/2} \cos^4 x \cdot \sin^3 x \, dx$$

Question 7 (Section B — Q8)

(a) Show the reduction formula for:

$$I = \int_0^{\pi/4} \tan^n x \, dx$$

(b) Use the reduction formula to evaluate:

$$\int x^e \cdot e^{-x} \, dx$$

— Wish you Best of Luck —